

	Course Name:	Applied Calculus	Course Code:	MT1001
	Program:	BEE	Semester:	Fall 2023
	Duration:	3 hours	Total Marks:	100
	Exam Date:	27 th December, 2023	Weight:	50
	Section:	ALL	Page(s):	2
	Exam Type:	Final Exam		

Student Name: _____ Roll No. _____ Section: _____

Instruction/Notes: 1. All CLOs are covered in this paper
2. If need arises, make valid assumptions and clearly mention it with your answer.

Question 1 [10 marks, CLO1]

- a. [3 marks] Illustrate the given inequality in terms of intervals and find the solution set on real line.
 $2x + 7 > 3.$
- b. [3 marks] Illustrate the given inequality in terms of intervals and find the solution set on real line.
 $x^2 + x > 1.$
- c. [4 marks] Identify whether the given function is even, odd, or neither and also sketch the graph of the function $y = |\sqrt{x} - 3|$, not by plotting points, but by starting with the graph of one of the standard function.

Question 2 [10 marks, CLO2]

- a. [5 marks] Determine the numbers at which f is discontinuous. At which of these numbers is f continuous from the right, from the left, or neither? Sketch the graph of f .

$$f(x) = \begin{cases} x^2 - 4, & x < 1 \\ 1, & x = 1 \\ -\frac{1}{2}x + 1, & x > 1 \end{cases}$$

- b. [5 marks] Evaluate the followings. If the symbol $[[]]$ denotes the greatest integer function.

(i) $\lim_{x \rightarrow -3.1} [[x]]$ (ii) $\lim_{x \rightarrow -3.9} [[x]]$ (iii) $\lim_{x \rightarrow 4.5} [[x]]$

Question 3 [20 marks, CLO3]

- a. [10 marks] Compute the derivative of the given function and sketch the curve by using differentiation
 $y = x^3 + 3x^2$
- b. [10 marks] Air is being pumped into a spherical balloon so that its volume increases at a rate of $100 \text{ cm}^3/\text{s}$. How fast is the radius of the balloon increasing when the diameter is 50 cm ?

Question 4 [40 marks, CLO4]

- a. [30 marks] Integrate the given functions by using any appropriate methods.

(i). $\frac{1}{2 \sin^2 x + 3 \cos^2 x}$

(ii). $\frac{\sec x}{1 + \csc x}$

(iii). $\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2 + 9)^{\frac{3}{2}}} dx.$

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b. [10 marks] Sketch the region enclosed by the given curves and find its area.

$$y = x^2, \quad y = 2x - x^2.$$

Question 5 [20 marks, CLO5]

(a). [5 marks] Write a vector equation, parametric equations and symmetric equation for the line. The line through the point $(0, 14, -10)$ and parallel to the line $x = -1 + 2t, y = 6 - 3t, z = 3 + 9t$.

(b). [5 marks] Formulate the distance between the given planes, if they are parallel.

$$2x - 3y + z = 4, \quad 4x + 2z - 6y = 3.$$

(c). [10 marks] Write an equation of the plane. The plane that passes through the point $(3, 1, 4)$ and contains the line of intersection of the planes $x + y + 3z = 1$ and $2x - y + z = -3$.